

Linux Cheat Sheet

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Linux File Structure

`/boot`	Boot loader files
`/bin`	executable binaries
`/home`	Users home directory
`/etc`	system config files
`/var/log`	log files
`/usr/local`	software manualy compiled or installed manually
`/dev/null`	when data is sent to this file, the data disappears.

Path shortcute

`~`	home directory
`/`	root directory
`.`	current directory
`..`	parent directory
`-`	previous directory

Linux Terminal

General Commands

`clear`	Clear the terminal screen.
`history`	Display the command history.
`cal`	Display a calendar.
`uptime`	Display how long the system has been running.
`man`	Display the manual for a command. (Tap q to quit)
`sleep 10`	Wait 10 seconds before stopping.
`yes hello`	print `hello` until the program is killed.
`cmd &`	run `cmd` in the background.
`date`	Display the current date and time.
`date "+Y-m-d 00:00:00"`	set the date and time.

File navigation and editing

`cd`	Change directory.
`ls`	List files in the current directory.
`ls -l`	List files with detailed information.
`ls -a`	List all files, including hidden ones.
`pwd`	Print the current working directory.
`open`	Open a file with the default application.
`mkdir`	Create a new directory.
`rm`	Remove a file.
`rm -rf`	Remove a directory and its contents.
`cp`	Copy a file or directory.
`mv`	Move or rename a file or directory.
`touch`	Create a new empty file.
`cat`	Display the contents of a file.
`more`	`cat` for big files.
`tail`	Display the last few lines of a file.
`tail -f`	Continuously display new lines added to a file.
`head`	Display the first few lines of a file.
`stat`	Display detailed information about a file.

To search the command history press `Ctrl + r`.

OS Commands

`sudo`	Execute a command with superuser privileges.
`sudo -l`	List permitted commands.
`whoami`	Display the current user.
`who`	Display all logged-in users.
`id`	Display user ID and group ID information.
`id -u`	Display the user ID.
`id -g`	Display the group ID.
`nproc`	show the number of processing units (CPU cores).
`ps`	Display currently running processes.
`top`	Show the current processes running alongside the cpu and ram usage.
`kill`	Terminate a process by its PID.
`kill -9`	Forcefully terminate a process by its PID.
`kill -15`	terminate and clean a process by its PID.
`reboot`	Restart the system.
`poweroff`	Shut down the system.
`useradd`	Add a new user.
`userdel`	Delete a user.
`passwd`	Change a user's password.
`su - user`	Switch to another user.
`chown user file`	Change the owner of a file to `user`.
`chown user:group file`	Change the owner and group of a file.
`chmod flag file -u tom`	Change the permissions of `file` for user `tom`.
`chmod +x file`	Add execute permission to a file.
`chmod +r file`	Add read permission to a file.
`chmod +w file`	Add write permission to a file.
`chmod -x file`	Remove execute permission from a file.
`chmod -r file`	Remove read permission from a file.
`chmod -w file`	Remove write permission from a file.

An alternative way of shutting down or reboot is to use `shutdown [OPTION] [TIME] [MESSAGE]` where `MESSAGE` shows a message to all users.

[TIME]	
`now`	run shutdown instantly.
`+m`	run shutdown after `m` minutes.
`hh:mm`	run shutdown at the absolute time `hh:mm`.
[OPTION]	
`-P`	This is the default option.
`-H`	Halt
`-r`	reboot
`--no-wall`	shutdown without warning.

Daemons

Daemons are a type of process.

Systemd

Systemd is the most widely used init system and service manager in modern Linux distributions. It is used in distributions such as Ubuntu, Debian, Fedora, and many others.

`systemctl start service`	Start a service.
`systemctl stop service`	Stop a service.
`systemctl restart service`	Restart a service.
`systemctl status service`	Check the status of a service.
`systemctl enable service`	Enable a service to start on boot.
`systemctl disable service`	Disable a service from starting on boot.
`systemctl daemon-reload`	Reload the systemd manager configuration
`systemctl reset-failed service`	Reset the failed state of a service.
`journalctl -u service`	View logs for a specific service.
`journalctl -n 100 -u service`	View the last 100 log entries for a specific service.
`--no-pager`	Disable paging of log output.

OpenRC

OpenRC is an alternative init system and service manager used in some Linux distributions, such as Alpine Linux and Gentoo.

`rc-service myService start`	Start a service.
`rc-service myService stop`	Stop a service.
`rc-service myService restart`	Restart a service.
`rc-service myService status`	Check the status of a service.
`rc-update add myService default`	Enable a service to start on boot.
`rc-update del myService default`	Disable a service from starting on boot.

Shell Scripting

operators

>	Redirect output to a file, overwriting the file if it exists.
>>	Redirect output to a file, appending to the file if it exists.
<	Redirect input from a file.
|	Pipe the output of one command to another command.
&&	Execute the next command only if the previous command succeeded.

Basic Syntax

`#!/bin/bash`	Shebang line to specify the script interpreter.
`echo`	Print text to the terminal.
`read`	Read input from the user.
`if`	Start of a conditional statement.
`elif`	Else if condition in a conditional statement.
`else`	Else condition in a conditional statement.
`fi`	End of a conditional statement.
`for`	Start of a for loop.
`while`	Start of a while loop.
`do`	Start of the loop body.
`done`	End of a loop.
`$0`	The name of the script.
`$1, $2, ...`	Positional parameters for script arguments.
`grep`	Search for a pattern in a file or input.
`exec`	Execute a command.
`exit`	Exit the script.

Networking Commands

The below commands requires the `net-tools` package to be installed on the system.

`ifconfig`	Displays IP address.
`ping`	Returns packet loss between two the device and a destination.
`traceroute`	Traces the path packets take to a destination.
`nslookup`	Returns IP address information for a domain name.
`arp -a`	Displays the ARP table, which maps IP addresses to MAC addresses.
`arp myIPa`	Displays the ARP table for the specific IP address.
`curl`	Transfers data from or to a server, using various protocols.
`curl ifconfig.me`	Display machine's public IPv4 address.

SSH connection

`ssh -i ~/.ssh/myKey user@ipv4`	Connect to a remote host via SSH.
`exit`	Exit the SSH session.
`Ctrl + D`	Exit the SSH session.

nginx

`nginx` is a software used to run a server on linux.

Cron

Cron is a time-based job scheduler. It allows users to schedule tasks (commands or scripts) to run automatically at specified intervals.

Basics

`crontab -l`	List the current user's crontab.
`crontab -e`	Edit the current user's crontab.
`crontab -r`	Remove the current user's crontab.
`crontab filename`	Install a new crontab from a file.
`crond`	The cron daemon sometime also called `cron`.

Syntax

`* * * * * command`	Run `command` every minute.
`0 * * * * command`	Run `command` at the start of every hour.
`0 0 * * * command`	Run `command` at midnight every day.
`0 0 * * 0 command`	Run `command` at midnight every Sunday.
`0 0 1 * * command`	Run `command` at midnight on the first day of every mont
`0 0 1 1 * command`	Run `command` at midnight on January 1st every year.

After the time and date fields, you can specify the command to be executed (note that cron dont use $PATH so you need to use absolute paths). Example:

```
[Desktop Entry]
Name=Outlook
Comment=Outlook in Chromium
Exec=chromium-browser --app=https://outlook.office.com/mail/
Icon=internet-mail
Type=Application
Categories=Network;Email;
StartupWMClass=outlook.office.com__mail
```

This will run the Python script `myscript.py` at midnight every Sunday and reboot the system at 5 AM every day.

package managers

`apt`	Debian-based(include ubuntu) package manager.
`apk`	Alpine Linux package manager.
`yum`	Red Hat-based package manager.
`dnf`	Fedora-based package manager.
`pacman`	Arch Linux package manager.
`brew`	Alternative package manager for macOS and many Linux distributions.

apk

`update`	Update repository indexes.
`upgrade`	Install upgrades available from repositories .
`list`	List all available packages.
`add pkg`	Install a package named `pkg`.
`del pkg`	Remove a package named `pkg`.
`search pkg`	Search for a package named `pkg`.
`info pkg`	Display detailed information about a package named `pkg`.

apt

`update`	Update the package list.
`upgrade`	Upgrade all installed packages to their latest versions.
`list`	List all available packages.
`list --upgradable`	List all packages that can be upgraded.
`install pkg`	Install a package named `pkg`.
`remove pkg`	Remove a package named `pkg`.
`search pkg`	Search for a package named `pkg`.

brew

`update`	Update the package list.
`upgrade`	Upgrade all installed packages to their latest versions.
`list`	List all available packages.
`install pkg`	Install a package named `pkg`. `--only-dependencies` self explenatory.
`uninstall pkg`	Remove a package named `pkg`.
`bundle install pkg`	Install a package named `pkg` and its dependencies.

Tips

Render a website as an app

If a app is not available for linux but can be used on a web browser (eg: outlook, spotify, etc.). you can use the command `chromium-browser --app=siteUrl` (eg: `chromium-browser --app=https://outlook.office365.com/mail`). Create a file `~/.local/share/applications/outlook.desktop` with the following inside:

```
[Desktop Entry]
Name=Outlook
Comment=Outlook in Chromium
Exec=chromium-browser --app=https://outlook.office.com/mail/
Icon=internet-mail
Type=Application
Categories=Network;Email;
StartupWMClass=outlook.office.com__mail
```